

“TAAT SHILPO” CHAINS FOR EFFICIENCY

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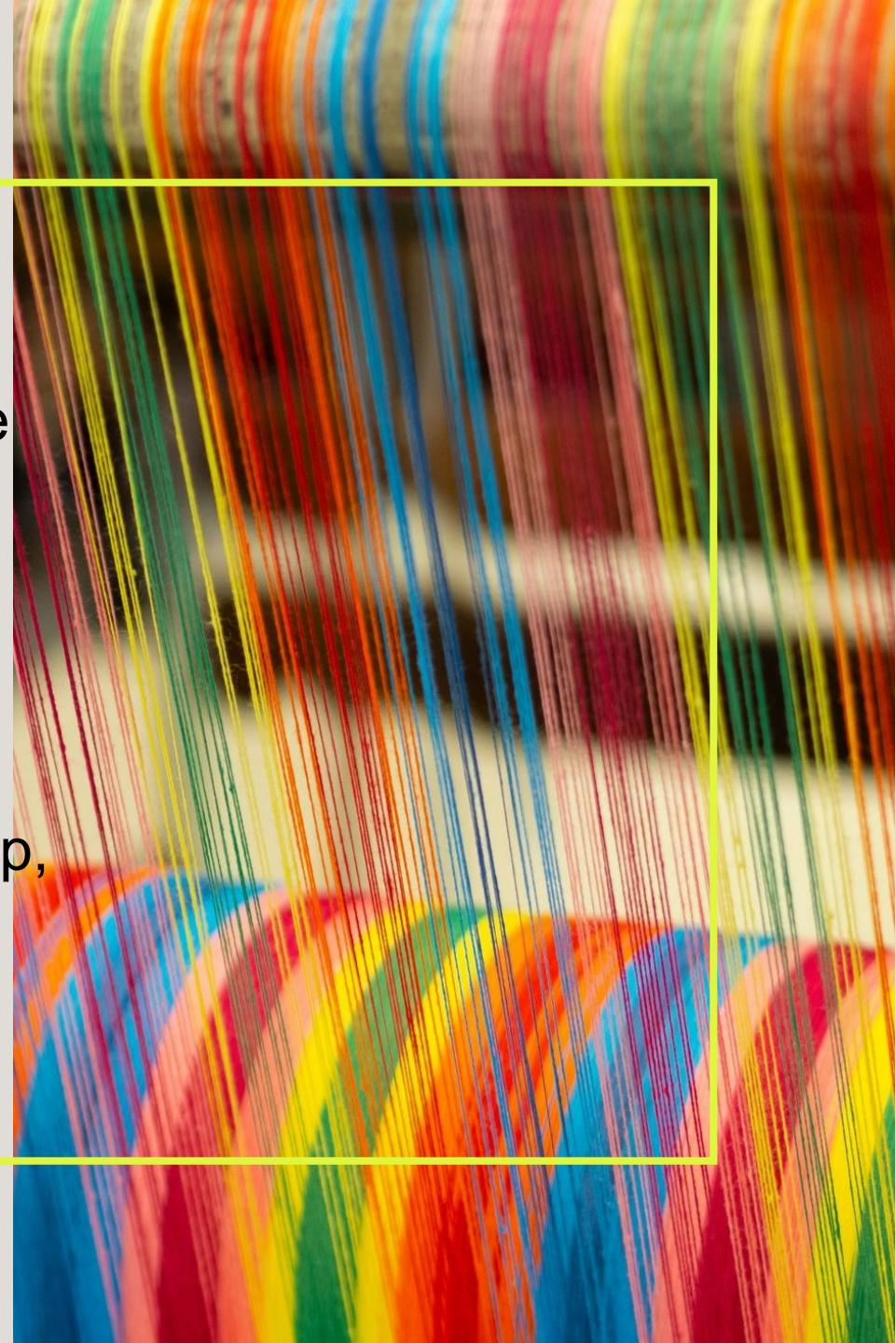
INTRODUCTION

- **Taat Shilpo Braiding**

Taat brading is a traditional handloom brading technique originating from specifically Bangladesh and the Indian state of West Bengal. It involves the intricate art of weaving decorative motifs directly onto the fabric, typically using cotton. The process of creating Taat Fabric is highly labor-intensive and requires exceptional skill and precision. It is known for their fine craftsmanship, intricate designs, and lightweight feel, making them average prized.

Project Objectives:

To design a database system to implement a systematic approach to streamline the various stages of Taat fabric production, including raw material sourcing, weaving, designing, marketing, and distribution.



Benefits of a Database Model for Taat Shilpo

In Bangladesh, most of the cases there is no systematic approach of producing and distributing this traditional saree in. Therefore, implementing a database model into Taat braiding of Bangladesh can bring several benefits such as:

1.Improved
inventory
management,

2.Design
preservation,

3.Efficient order
tracking,

4.Enhanced
quality control,

5.Cost
optimization,

6.Customer
Relationship
Management
(CRM)

7.Data-driven
decision-making,

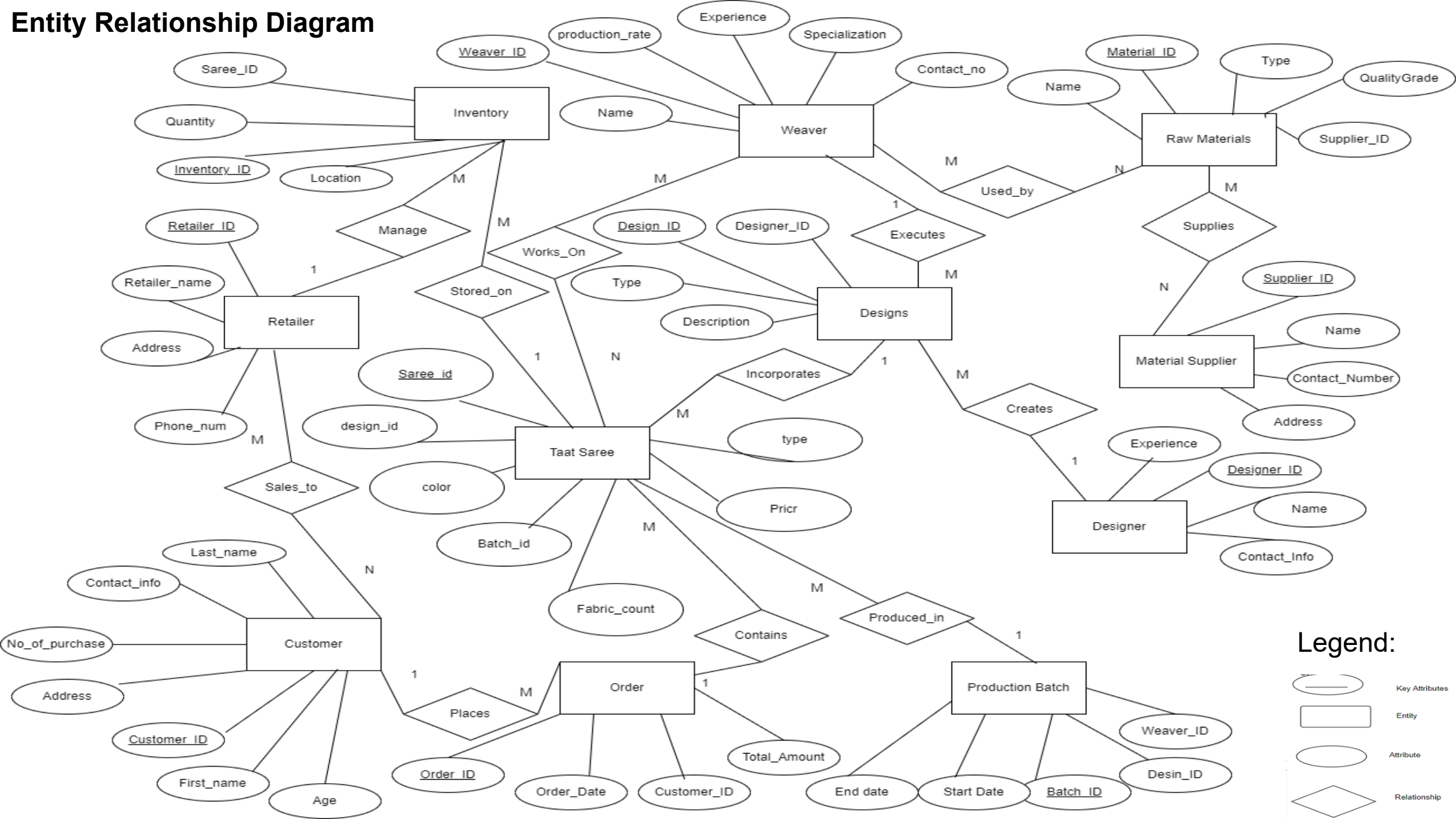
8.Streamlined
workflow



Entities & Attributes

1. Material Supplier : SupplierID, Name, Contact_Number, Address
2. Raw Materials: MaterialID, Type, QualityGrade, WeaverID
3. Weavers: WeaverID, Name, ContactNumber, Specialization, Experience, ProductionRate
4. Designer: DesignerID, Name,ContactInfo, Experience
5. Designs: DesignID, Type, Description, DesignerID
6. Production Batch: BatchID, StartDate, EndDate, DesignID , WeaverID
7. Taat Saree: SareeID, Type, Color,Price, FabricCount, BatchID
8. Retailer :RetailerID, RetailerName, Phone_Num,Address 9. Customer:CustomerID, FirstName, LastName, ContactInfo, Address, Age, NoOfPurchases
10. Inventory : InventoryID, Quantity, Location, SareeID 11.Order :OrderID, CustomerID ,OrderDate, TotalAmount

Entity Relationship Diagram



Relationship Instances:

1. Material Supplier supplies Raw Materials (1:M)
2. Raw Materials are used by weaver(M:N)
3. Weaver works on Taat Saree(M:N)
4. Taat Saree incorporates design(M:1)
5. Designer creates Designs (1: M)
6. Taat Saree contains order(M:1)
7. Taat Saree stores on Inventory (1:M)
8. Retailer manages Inventory (1:M)
9. Retailer sells to Customer (M:N)
10. Customer places Order(1:M)

Create Commands(Cont.)

Create MaterialSupplier table:

```
CREATE TABLE MaterialSupplier (  
SupplierID INT PRIMARY KEY, Name  
VARCHAR(255), Contact_Number  
VARCHAR(255), Address  
VARCHAR(255));
```

Create Designer table:

```
CREATE TABLE Designer (  
DesignerID INT PRIMARY KEY,  
Name VARCHAR(255),  
ContactInfo VARCHAR(255),  
Experience INT);
```

Create 'Order' table:

```
CREATE TABLE `Order` (OrderID  
INT PRIMARY KEY, CustomerID  
INT, OrderDate DATE,  
TotalAmount DECIMAL(10,2),  
FOREIGN KEY (CustomerID)  
REFERENCES  
Customer(CustomerID));
```

Create Inventory table:

```
CREATE TABLE Inventory (  
InventoryID  
INT PRIMARY KEY, SareeID INT,  
Quantity INT, Location VARCHAR(255),  
FOREIGN KEY (SareeID)  
REFERENCES TaatSaree(SareeID));
```

Create Designs table:

```
CREATE TABLE Designs (  
DesignID INT PRIMARY KEY,  
Type VARCHAR(255),  
Description TEXT, DesignerID  
INT, FOREIGN KEY  
(DesignerID) REFERENCES  
Designer(DesignerID));
```

Create Production Batch table

```
CREATE TABLE  
ProductionBatch (  
BatchID INT PRIMARY KEY,  
StartDate DATE, EndDate  
DATE, DesignID INT,  
WeaverID INT, FOREIGN KEY  
(DesignID) REFERENCES  
Designs(DesignID),  
FOREIGN KEY (WeaverID)  
REFERENCES  
Weaver(WeaverID));
```


Create Commands(Cont.)

Create Retailer table:

```
CREATE TABLE Retailer ( RetailerID  
INT PRIMARY KEY, RetailerName  
VARCHAR(100), Address  
VARCHAR(255), Phone_Num  
VARCHAR(255));
```

Create Weaver Table

```
CREATE TABLE Weaver (  
WeaverID INT PRIMARY KEY,  
Name VARCHAR(100),  
ContactNumber VARCHAR(15),  
Specialization VARCHAR(100),  
ProductionRate DECIMAL(10, 2),  
Experience INT);
```

Create Raw Materials Table:

```
1.CREATE TABLE RawMaterials (  
MaterialID INT PRIMARY KEY, Name  
VARCHAR(255), Type  
VARCHAR(255), QualityGrade  
VARCHAR(255), SupplierID INT,  
FOREIGN KEY (SupplierID)  
REFERENCES  
MaterialSupplier(SupplierID));
```

Create Customer Table:

```
CREATE TABLE Customer (  
CustomerID INT PRIMARY  
KEY, FirstName  
VARCHAR(50), LastName  
VARCHAR(50), ContactNum  
VARCHAR(15), Address  
VARCHAR(255),  
PurchaseHistory  
VARCHAR(255),  
LastDateOfPurchase DATE);
```

Create Taat Saree Table:

```
CREATE TABLE TaatSaree ( SareeID  
INT PRIMARY KEY, Type  
VARCHAR(255), Color VARCHAR(50),  
Price DECIMAL(10, 2), FabricCount  
INT, DesignID INT, BatchID INT,  
FOREIGN KEY (DesignID)  
REFERENCES Designs(DesignID),  
FOREIGN KEY (BatchID)  
REFERENCES  
BatchID(BatchID));
```

MariaDB [19980ak]> select * from MaterialSupplier;

SupplierID	Name	Contact_Number	Address
1	ABC Supplies	+8801724632658	123 Main St, Dhaka
2	WeaveMaster Co.	+8801724632659	456 Weave Lane, Narayanganj
3	ThreadWorks	+8801724632660	789 Fabric Road, Chittagong
4	Golden Fibers	+8801724632661	101 Silk Street, Khulna
5	CraftSource	+8801724632662	202 Artisan Ave, Rajshahi
6	Loom & Thread	+8801724632663	303 Cotton Blvd, Sylhet
7	FiberKing	+8801724632664	404 Yarn Parkway, Barisal
8	Elegant Textiles	+8801724632665	505 Textile Plaza, Gazipur
9	Premium Yarns	+8801724632666	606 Weavers Way, Comilla
10	Handloom Hub	+8801724632667	707 Artisan Alley, Mymensingh

MariaDB [19980ak]> select * from Designer;

DesignerID	Name	ContactInfo	Experience
1	Rahim Uddin	+8801711122233	10
2	Karim Ahmed	+8801722233344	8
3	Ayesha Siddiqua	+8801733344455	12
4	Fatima Begum	+8801744455566	15
5	Nasima Khatun	+8801755566677	7
6	Anisur Rahman	+8801766677788	9
7	Shafiqur Rahman	+8801777788899	14
8	Rashida Akhtar	+8801788899900	11
9	Tasnim Akhter	+8801799900111	6
10	Mahmudul Hassan	+8801700001222	13

MariaDB [19980ak]> select * from Customer;

CustomerID	FirstName	LastName	ContactNum	Address	PurchaseHistory	LastDateOfPurchase
101	Rahim	Uddin	+8801711122233	123 Elm St, Dhaka	Saree A, Saree B	2023-12-01
102	Karim	Ahmed	+8801722233344	456 Oak St, Narayanganj	Saree C	2023-12-02
103	Ayesha	Siddiqua	+8801733344455	789 Pine St, Chittagong	Saree D, Saree E	2023-12-03
104	Fatima	Begum	+8801744455566	101 Maple St, Khulna	Saree F	2023-12-04
105	Nasima	Khatun	+8801755566677	202 Birch St, Rajshahi	Saree G, Saree H	2023-12-05
106	Anisur	Rahman	+8801766677788	303 Cedar St, Sylhet	Saree I	2023-12-06
107	Shafiqur	Rahman	+8801777788899	404 Walnut St, Barisal	Saree J	2023-12-07
108	Rashida	Akhtar	+8801788899900	505 Chestnut St, Gazipur	Saree K	2023-12-08
109	Tasnim	Akhter	+8801799900111	606 Willow St, Comilla	Saree L	2023-12-09
110	Mahmudul	Hassan	+8801700001222	707 Spruce St, Mymensingh	Saree M	2023-12-10

MariaDB [19980ak]> SELECT * FROM `Order`;

OrderID	CustomerID	OrderDate	TotalAmount
1	101	2024-01-15	2500.00
2	102	2024-01-18	3000.50
3	103	2024-01-20	1500.75
4	104	2024-01-22	3200.00
5	105	2024-01-25	4500.25
6	106	2024-01-28	2750.00
7	107	2024-02-01	2900.10
8	108	2024-02-05	3600.60
9	109	2024-02-10	4100.80
10	110	2024-02-15	2200.90

MariaDB [19980ak]> select * from RawMaterials;

MaterialID	Name	Type	QualityGrade	SupplierID
1	Cotton Taat	Fabric	A	1
2	Silk Taat	Fabric	A	2
3	Golden Thread	Thread	A	3
4	Silver Thread	Thread	B	4
5	Organic Dye	Color	B	5
6	Organic Cotton Taat	Fabric	A	1
7	Synthetic Fiber Taat	Fabric	C	2
8	Natural Dye	Color	A	3
9	Polyester Taat	Fabric	B	4
10	Linen Taat	Fabric	A	5

MariaDB [19980ak]> select * from Weaver;

WeaverID	Name	ContactNumber	Specialization	ProductionRate	Experience
1	Abdul Karim	+8801712345678	Cotton Weaving	20.50	10
2	Rahima Begum	+8801723456789	Silk Weaving	18.75	8
3	Jamal Uddin	+8801734567890	Golden Thread Weaving	22.30	12
4	Hasina Khatun	+8801745678901	Silver Thread Weaving	19.20	9
5	Mizanur Rahman	+8801756789012	Organic Cotton Weaving	21.00	11
6	Selina Akhtar	+8801767890123	Synthetic Fiber Weaving	17.85	7
7	Nurul Islam	+8801778901234	Natural Dyeing	23.40	13
8	Fatema Banu	+8801789012345	Polyester Weaving	20.00	10
9	Habibur Rahman	+8801790123456	Linen Weaving	19.50	8
10	Sufia Begum	+8801701234567	Mixed Fabric Weaving	18.25	6


```
MariaDB [19980ak]> select * from Designs;
```

DesignID	Type	Description	DesignerID
1	Traditional Taat	A classic Taat design featuring intricate geometric patterns.	1
2	Modern Taat	A contemporary take on Taat with bold, abstract motifs.	2
3	Floral Taat	Taat fabric with delicate floral designs and motifs.	3
4	Heritage Taat	A design inspired by traditional Bangladeshi heritage.	4
5	Eco-friendly Taat	Sustainable Taat designs using organic materials.	5
6	Festive Taat	Bright and vibrant Taat patterns suitable for festivals.	6
7	Casual Taat	Simple and elegant Taat designs for everyday wear.	7
8	Luxury Taat	High-end Taat with intricate details and premium materials.	8
9	Minimalist Taat	Clean and minimal Taat designs with subtle patterns.	9
10	Vintage Taat	A retro-inspired Taat design with classic elements.	10

```
MariaDB [19980ak]> select * from ProductionBatch;
```

BatchID	StartDate	EndDate	DesignID	WeaverID
2001	2024-02-01	2024-03-15	1	1
2002	2024-03-16	2024-06-30	2	2
2003	2024-01-01	2024-05-15	3	3
2004	2023-11-16	2024-04-30	4	4
2005	2023-12-01	2024-05-15	5	5
2006	2024-02-16	2024-07-30	6	6
2007	2024-06-01	2024-08-15	7	7
2008	2023-10-16	2024-04-30	8	8
2009	2024-07-01	2024-09-15	9	9
2010	2024-08-16	2024-11-30	10	10

MariaDB [19980ak]> select * from Retailer;

RetailerID	RetailerName	Address	Phone_Num
1	Dhaka Saree House	123 Dhaka St, Dhaka	+8801712345678
2	Chittagong Textiles	456 Chittagong Ave, Chittagong	+8801723456789
3	Rajshahi Handloom	789 Rajshahi Blvd, Rajshahi	+8801734567890
4	Khulna Weaves	101 Khulna Road, Khulna	+8801745678901
5	Sylhet Silk Shop	202 Sylhet Lane, Sylhet	+8801756789012
6	Barisal Fabric Store	303 Barisal Path, Barisal	+8801767890123
7	Gazipur Garments	404 Gazipur Way, Gazipur	+8801778901234
8	Comilla Couture	505 Comilla St, Comilla	+8801789012345
9	Narayanganj Threads	606 Narayanganj Ave, Narayanganj	+8801790123456
10	Mymensingh Looms	707 Mymensingh Blvd, Mymensingh	+8801701234567

MariaDB [19980ak]> select * from TaatSaree;

SareeID	Type	Color	Price	FabricCount	DesignID	BatchID
1	Traditional	Red	1500.00	4	1	2001
2	Modern	Blue	1600.00	5	2	2002
3	Floral	Green	1700.00	6	3	2003
4	Heritage	Yellow	1800.00	7	4	2004
5	Eco-friendly	White	1900.00	8	5	2005
6	Festive	Pink	2000.00	9	6	2006
7	Casual	Orange	2100.00	10	7	2007
8	Luxury	Purple	2200.00	11	8	2008
9	Minimalist	Grey	2300.00	12	9	2009
10	Vintage	Black	2400.00	13	10	2010

MariaDB [19980ak]> select * from Inventory;

InventoryID	SareeID	Quantity	Location
1	1	50	Dhaka Warehouse
2	2	40	Chittagong Warehouse
3	3	30	Rajshahi Warehouse
4	4	20	Khulna Warehouse
5	5	60	Sylhet Warehouse
6	6	45	Barisal Warehouse
7	7	35	Gazipur Warehouse
8	8	25	Comilla Warehouse
9	9	55	Narayanganj Warehouse
10	10	65	Mymensingh Warehouse

BASIC COMMAND



1. Retrieve all RawMaterials with QualityGrade 'A'

SELECT * FROM RawMaterials WHERE QualityGrade =

```
MariaDB [19980ak]> SELECT * FROM RawMaterials WHERE QualityGrade = 'A';
```

MaterialID	Name	Type	QualityGrade	SupplierID
1	Cotton Taat	Fabric	A	1
2	Silk Taat	Fabric	A	2
3	Golden Thread	Thread	A	3
6	Organic Cotton Taat	Fabric	A	1
8	Natural Dye	Color	A	3
10	Linen Taat	Fabric	A	5

Mathematical Notation

$\sigma_{\text{QualityGrade}='A'}(\text{RawMaterials})$

This command helps in identifying high-quality materials, ensuring that only the best raw materials are used in the production of Taat sarees.

2. Retrieve all customers whose last name starts with the letter 'A'

```
SELECT CustomerID, FirstName, LastName,
ContactNum
FROM Customer
WHERE LastName LIKE 'A%';
```

```
MariaDB [19980ak]> SELECT CustomerID, FirstName, LastName, ContactNum
-> FROM Customer
-> WHERE LastName LIKE 'A%';
```

CustomerID	FirstName	LastName	ContactNum
102	Karim	Ahmed	+8801722233344
108	Rashida	Akhtar	+8801788899900
109	Tasnim	Akhter	+8801799900111

This command helps in retrieving the details of customers whose last names start with the letter 'A'. It enables targeted communication and marketing efforts by focusing on a specific group of customers based on their last names.

Mathematical Notation

$$\pi_{CustomerID,FirstName,LastName,ContactNum}(\sigma_{LastName \text{ LIKE 'A\%'}}(Customer))$$

3. Retrieve the total number of sarees grouped by color

```
SELECT Color, COUNT(SareeID) AS NumberOfSarees
FROM TaatSaree
GROUP BY Color;
```

```
MariaDB [19980ak]> SELECT Color, COUNT(SareeID) AS NumberOfSarees
-> FROM TaatSaree
-> GROUP BY Color;
```

Color	NumberOfSarees
Black	1
Blue	1
Green	1
Grey	1
Orange	1
Pink	1
Purple	1
Red	1
White	1
Yellow	1

This command helped in understanding the distribution of sarees across different colors, providing insights into which colors are more popular or in higher demand.

Mathematical Notation:

$$\pi_{Color, COUNT(SareeID) \rightarrow NumberOfSarees}(\gamma_{Color}(Taatsaree))$$

4. Retrieve saree details ordered by price in descending order

```
SELECT SareeID, Type, Color, Price
FROM TaatSaree
ORDER BY Price DESC;
```

```
MariaDB [19980ak]> SELECT SareeID, Type, Color, Price
-> FROM TaatSaree
-> ORDER BY Price DESC;
```

SareeID	Type	Color	Price
10	Vintage	Black	2400.00
9	Minimalist	Grey	2300.00
8	Luxury	Purple	2200.00
7	Casual	Orange	2100.00
6	Festive	Pink	2000.00
5	Eco-friendly	White	1900.00
4	Heritage	Yellow	1800.00
3	Floral	Green	1700.00
2	Modern	Blue	1600.00
1	Traditional	Red	1500.00

This command facilitated the identification of the most expensive sarees, aiding in pricing strategy and inventory management by highlighting high-value items.

Mathematical Notation:

$$\pi_{SareeID, Type, Color, Price}(\tau_{Price \text{ DESC}}(Taatsaree))$$

5. List the average price of sarees grouped by color having an average price over \$180

```
SELECT Color, AVG(Price) AS AvgPrice
FROM TaatSaree
GROUP BY Color
HAVING AVG(Price) > 180;
```

```
MariaDB [19980ak]> SELECT Color, AVG(Price) AS AvgPrice
-> FROM TaatSaree
-> GROUP BY Color
-> HAVING AVG(Price) > 180;
```

Color	AvgPrice
Black	2400.000000
Blue	1600.000000
Green	1700.000000
Grey	2300.000000
Orange	2100.000000
Pink	2000.000000
Purple	2200.000000
Red	1500.000000
White	1900.000000
Yellow	1800.000000

This command helps in understanding the pricing trends based on saree colors, assisting in strategic pricing and inventory decisions.

Mathematical Notation

$$\pi_{\text{Color, AvgPrice}}(\sigma_{\text{AvgPrice} > 180}(\gamma_{\text{Color; AvgPrice} = \text{AVG}(\text{Price})}(\text{Taatsaree})))$$

INTERMEDIATE COMMANDS



1. List the total quantity of each type of RawMaterial supplied by each MaterialSupplier

```
SELECT ms.Name, rm.Type, COUNT(rm.MaterialID) AS  
TotalMaterials  
FROM MaterialSupplier ms  
JOIN RawMaterials rm ON ms.SupplierID = rm.SupplierID  
GROUP BY ms.Name, rm.Type;
```

```
MariaDB [19980ak]> SELECT ms.Name, rm.Type, COUNT(rm.MaterialID) AS TotalMaterials  
-> FROM MaterialSupplier ms  
-> JOIN RawMaterials rm ON ms.SupplierID = rm.SupplierID  
-> GROUP BY ms.Name, rm.Type;
```

Name	Type	TotalMaterials
ABC Supplies	Fabric	2
CraftSource	Color	1
CraftSource	Fabric	1
Golden Fibers	Fabric	1
Golden Fibers	Thread	1
ThreadWorks	Color	1
ThreadWorks	Thread	1
WeaveMaster Co.	Fabric	2

$\pi_{ms.Name, rm.Type, TotalMaterials}(\gamma_{ms.Name, rm.Type, TotalMaterials=COUNT(rm.MaterialID)}(MaterialSupplier \bowtie RawMaterials))$

This command retrieves the total quantity of each type of raw material supplied by each supplier. It helps in understanding the contribution of each supplier to the inventory of raw materials, aiding in supplier management and procurement planning.

2.Retrieve the names of all Weavers and the total number of sarees they have worked on, grouped by Weaver

```
SELECT w.Name, COUNT(ts.SareeID) AS TotalSareesWorkedOn
FROM Weaver w
JOIN ProductionBatch pb ON w.WeaverID = pb.WeaverID
JOIN TaatSaree ts ON pb.BatchID = ts.BatchID
GROUP BY w.Name;
```

```
MariaDB [19980ak]> SELECT w.Name, COUNT(ts.SareeID) AS TotalSareesWorkedOn
-> FROM Weaver w
-> JOIN ProductionBatch pb ON w.WeaverID = pb.WeaverID
-> JOIN TaatSaree ts ON pb.BatchID = ts.BatchID
-> GROUP BY w.Name;
```

Name	TotalSareesWorkedOn
Abdul Karim	1
Fatema Banu	1
Habibur Rahman	1
Hasina Khatun	1
Jamal Uddin	1
Mizanur Rahman	1
Nurul Islam	1
Rahima Begum	1
Selina Akhtar	1
Sufia Begum	1

Mathematical Notation

$\pi_{w.Name, TotalSareesWorkedOn}(\gamma_{w.Name; TotalSareesWorkedOn=COUNT(ts.SareeID)}((Weaver \bowtie_{WeaverID} ProductionBatch) \bowtie_{BatchID} TaatSaree))$

•This command provides the names of weavers along with the total number of sarees they have worked on.. It assists in evaluating the productivity of weavers and helps in identifying the most efficient weavers.

3. Retrieve the Retailer names and the number of TaatSarees they have in their inventory

```
SELECT i.Location, COUNT(i.SareeID) AS  
NumberOfSarees  
FROM Inventory i
```

```
MariaDB [19980ak]> SELECT i.Location, COUNT(i.SareeID) AS NumberOfSarees  
-> FROM Inventory i  
-> GROUP BY i.Location;
```

Location	NumberOfSarees
Barisal Warehouse	1
Chittagong Warehouse	1
Comilla Warehouse	1
Dhaka Warehouse	1
Gazipur Warehouse	1
Khulna Warehouse	1
Mymensingh Warehouse	1
Narayanganj Warehouse	1
Rajshahi Warehouse	1
Sylhet Warehouse	1

- This command lists the retailer names and the number of sarees they have in their inventory. It aids in inventory management by providing insights into the distribution of sarees across different retailers.

4. Retrieve the names of Customers and the total amount they have spent, grouped by Customer

```
SELECT c.FirstName, c.LastName, SUM(o.TotalAmount)
AS TotalSpent
FROM Customer c
JOIN `Order` o ON c.CustomerID = o.CustomerID
GROUP BY c.FirstName, c.LastName;
```

```
MariaDB [19980ak]> SELECT c.FirstName, c.LastName, SUM(o.TotalAmount) AS TotalSpent
-> FROM Customer c
-> JOIN `Order` o ON c.CustomerID = o.CustomerID
-> GROUP BY c.FirstName, c.LastName;
```

FirstName	LastName	TotalSpent
Anisur	Rahman	2750.00
Ayesha	Siddiqua	1500.75
Fatima	Begum	3200.00
Karim	Ahmed	3000.50
Mahmudul	Hassan	2200.90
Nasima	Khatun	4500.25
Rahim	Uddin	2500.00
Rashida	Akhtar	3600.60
Shafiqur	Rahman	2900.10
Tasnim	Akhter	4100.80

- This command retrieves the names of customers along with the total amount they have spent. It helps in understanding customer spending behavior,
- which is valuable for sales analysis and customer relationship management.

5. Retrieve the names of Designers and the number of designs they have created

```
SELECT d.Name, COUNT(des.DesignID)
AS NumberOfDesigns
FROM Designer d
JOIN Designs des ON d.DesignerID =
des.DesignerID
GROUP BY d.Name;
```

```
MariaDB [19980ak]> SELECT d.Name, COUNT(des.DesignID) AS NumberOfDesigns
-> FROM Designer d
-> JOIN Designs des ON d.DesignerID = des.DesignerID
-> GROUP BY d.Name;
```

Name	NumberOfDesigns
Anisur Rahman	1
Ayesha Siddiqua	1
Fatima Begum	1
Karim Ahmed	1
Mahmudul Hassan	1
Nasima Khatun	1
Rahim Uddin	1
Rashida Akhtar	1
Shafiqur Rahman	1
Tasnim Akhter	1

This command lists the names of designers and the number of designs they have created.

It helps in assessing the productivity and creativity of designers, which is essential for design and production planning.

**ADVANCED
COMMAND**



1. Retrieve the details of the top 3 most expensive sarees, including their type, color, price, designer's name, and the weaver who worked on them

```
SELECT ts.Type,
ts.Color, ts.Price,
d.Name AS
DesignerName,
w.Name AS
WeaverName
FROM TaatSaree ts
JOIN Designs des ON
ts.DesignID =
des.DesignID
JOIN Designer d ON
des.DesignerID =
d.DesignerID
JOIN ProductionBatch
pb ON ts.BatchID =
pb.BatchID
JOIN Weaver w ON
pb.WeaverID =
w.WeaverID
ORDER BY ts.Price
DESC
LIMIT 3;
```

```
MariaDB [19980ak]> SELECT ts.Type, ts.Color, ts.Price, d.Name AS DesignerName, w.Name AS WeaverName
-> FROM TaatSaree ts
-> JOIN Designs des ON ts.DesignID = des.DesignID
-> JOIN Designer d ON des.DesignerID = d.DesignerID
-> JOIN ProductionBatch pb ON ts.BatchID = pb.BatchID
-> JOIN Weaver w ON pb.WeaverID = w.WeaverID
-> ORDER BY ts.Price DESC
-> LIMIT 3;
```

Type	Color	Price	DesignerName	WeaverName
Vintage	Black	2400.00	Mahmudul Hassan	Sufia Begum
Minimalist	Grey	2300.00	Tasnim Akhter	Habibur Rahman
Luxury	Purple	2200.00	Rashida Akhtar	Fatema Banu

```
πts.Type, ts.Color, ts.Price, d.Name, w.Name
(σTop 3(γts.Type, ts.Color, ts.Price, d.Name, w.Name
((TaatSaree ⋈DesignID Designs) ⋈DesignerID Designer) ⋈BatchID
ProductionBatch) ⋈WeaverID Weaver)))
```

This advanced command helped identify the highest-value products and the artisans involved, supporting targeted marketing and showcasing premium offerings.

2. Retrieve the average price of sarees produced in each ProductionBatch and the name of the Weaver responsible, sorted by the highest average price

```
SELECT w.Name, pb.BatchID, AVG(ts.Price) AS AvgPrice
FROM Weaver w
JOIN ProductionBatch pb ON w.WeaverID = pb.WeaverID
JOIN TaatSaree ts ON pb.BatchID = ts.BatchID
GROUP BY w.Name, pb.BatchID
ORDER BY AvgPrice DESC;
```

```
MariaDB [19980ak]> SELECT w.Name, pb.BatchID, AVG(ts.Price) AS AvgPrice
-> FROM Weaver w
-> JOIN ProductionBatch pb ON w.WeaverID = pb.WeaverID
-> JOIN TaatSaree ts ON pb.BatchID = ts.BatchID
-> GROUP BY w.Name, pb.BatchID
-> ORDER BY AvgPrice DESC;
```

Name	BatchID	AvgPrice
Sufia Begum	2010	2400.000000
Habibur Rahman	2009	2300.000000
Fatema Banu	2008	2200.000000
Nurul Islam	2007	2100.000000
Selina Akhtar	2006	2000.000000
Mizanur Rahman	2005	1900.000000
Hasina Khatun	2004	1800.000000
Jamal Uddin	2003	1700.000000
Rahima Begum	2002	1600.000000
Abdul Karim	2001	1500.000000

This query provided insights into the productivity and value generated by each weaver, aiding in performance evaluation and incentive planning. It shows which weavers are involved in batches with higher average prices, the company can understand which weavers are more efficient or skilled in producing high-value sarees.

3. Retrieve the total number of sarees produced by each Designer, including the designer's name and sorted by the total number of sarees in descending order

```
SELECT d.Name AS DesignerName, COUNT(ts.SareeID) AS TotalSarees
FROM Designer d
LEFT JOIN Designs des ON d.DesignerID = des.DesignerID
LEFT JOIN TaatSaree ts ON des.DesignID = ts.DesignID
GROUP BY d.Name
ORDER BY TotalSarees DESC;
```

```
MariaDB [19980ak]> SELECT d.Name AS DesignerName, COUNT(ts.SareeID) AS TotalSarees
-> FROM Designer d
-> LEFT JOIN Designs des ON d.DesignerID = des.DesignerID
-> LEFT JOIN TaatSaree ts ON des.DesignID = ts.DesignID
-> GROUP BY d.Name
-> ORDER BY TotalSarees DESC;

+-----+-----+
| DesignerName | TotalSarees |
+-----+-----+
| Rahim Uddin  | 1           |
| Karim Ahmed  | 1           |
| Ayesha Siddiqua | 1          |
| Fatima Begum  | 1           |
| Nasima Khatun | 1           |
| Anisur Rahman | 1           |
| Shafiqur Rahman | 1          |
| Rashida Akhtar | 1           |
| Tasnim Akhter  | 1           |
| Mahmudul Hassan | 1          |
+-----+-----+
```

This command was useful for understanding the contribution of each designer to overall production, informing design and production strategy.

4. List Customers and Their Orders for Sarees Weaved in the Last 6 Months:

```
SELECT C.CustomerID, C.FirstName, C.LastName, O.OrderID, PB.StartDate
FROM Customer AS C
RIGHT JOIN `Order` AS O ON C.CustomerID = O.CustomerID
RIGHT JOIN TaatSaree AS S ON O.OrderID = S.SareeID
RIGHT JOIN (
    SELECT PB.BatchID, PB.StartDate
    FROM ProductionBatch AS PB
    WHERE PB.StartDate >= DATE_SUB(CURDATE(), INTERVAL 6 MONTH)
) AS PB ON S.BatchID = PB.BatchID;
```

```
MariaDB [19980ak]> SELECT C.CustomerID, C.FirstName, C.LastName, O.OrderID, PB.StartD
ate
```

```
-> FROM Customer AS C
-> RIGHT JOIN `Order` AS O ON C.CustomerID = O.CustomerID
-> RIGHT JOIN TaatSaree AS S ON O.OrderID = S.SareeID
-> RIGHT JOIN (
->     SELECT PB.BatchID, PB.StartDate
->     FROM ProductionBatch AS PB
->     WHERE PB.StartDate >= DATE_SUB(CURDATE(), INTERVAL 6 MONTH)
-> ) AS PB ON S.BatchID = PB.BatchID;
```

CustomerID	FirstName	LastName	OrderID	StartDate
102	Karim	Ahmed	2	2024-03-16
106	Anisur	Rahman	6	2024-02-16
107	Shafiqur	Rahman	7	2024-06-01
109	Tasnim	Akhter	9	2024-07-01
110	Mahmudul	Hassan	10	2024-08-16

This query helped track recent customer orders, ensuring timely follow-ups and enhancing customer relationship management.

5. Retrieve information about Jamdani sarees produced by experienced designers in batches, along with details of the production batch and the weaver involved.

```
SELECT S.*
FROM TaatSaree AS S
JOIN (
    SELECT De.DesignID
    FROM Designs AS De
    INNER JOIN Designer AS D ON De.DesignerID = D.DesignerID
    WHERE D.Experience > 5
) AS ExperiencedDesigns ON S.DesignID = ExperiencedDesigns.DesignID
JOIN ProductionBatch AS PB ON S.BatchID = PB.BatchID
JOIN Weaver AS W ON PB.WeaverID = W.WeaverID;
```

SareeID	Type	Color	Price	FabricCount	DesignID	BatchID
1	Traditional	Red	1500.00	4	1	2001
2	Modern	Blue	1600.00	5	2	2002
3	Floral	Green	1700.00	6	3	2003
4	Heritage	Yellow	1800.00	7	4	2004
5	Eco-friendly	White	1900.00	8	5	2005
6	Festive	Pink	2000.00	9	6	2006
7	Casual	Orange	2100.00	10	7	2007
8	Luxury	Purple	2200.00	11	8	2008
9	Minimalist	Grey	2300.00	12	9	2009
10	Vintage	Black	2400.00	13	10	2010

This command highlighted the collaboration between experienced designers and weavers, showcasing the craftsmanship and quality of Taat Shilpo's products.

- **Conclusion**

Taat Shilpo Supply Chain Efficiency Enhancement

- Structured database streamlines processes like raw material sourcing, production tracking, inventory management, and sales analysis.
- Facilitates better organization of data, enabling quick access to crucial information.
- Supports data-driven decision-making.
- Optimizes operational workflows, improves quality control, and manages cost.
- Data modeling modernizes and sustains traditional Taat Shilpo industry.

References

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[Fundamentals of Database Systems - Ramez Elmasri, Sham Navathe - Google Books](#)

THANK YOU

